

KrishiTech — Repository tree

Source: docs/project-structure.md §2 · Agentic agriculture platform for India · generated 2026-08-23

Phases: **P1** MVP **P1.5** fast-follow **P2** later · **✓ written** = doc complete

agri-os/

├─ CLAUDE.md

P1 Router: commands, read-before-you-build, non-negotiables, ask-don't-decide

├─ README.md

P1 Setup, run, test, deploy

├─ .env.example

P1 Required env vars, no secrets

├─ docs/

├─ README.md

P1 Doc index: what belongs in each file, and open questions

├─ project-structure.md

P1 **✓ written** – Repo tree, locked decisions, MVP scope, CI gates, open questions.

├─ sources.md

P1 Citations for asserted figures

├─ architecture.md

P1 Five layers, module map, scalability levers

├─ agent-contracts.md

P1 **✓ written** – uniform I/O, manifest, nine-section rule, agent catalog

├─ navigation-ia.md

P1 **✓ written** – 68 screens across three roles, states, backing agents

├─ design-system.md

P1 **✓ written** – two-layer tokens, role palettes, type scale per role, voice-state tokens

├─ offline-matrix.md

P1 **✓ written** – capability matrix, outbox, conflicts, media queue

├─ data-model.md

P1 Farm graph entities, event log, role projections, outcome ledger

├─ identity.md

P1 Phone-as-person, number change, shared devices, dormancy

├─ role-permissions.md

P1 Who reads/writes what; consent defaults; cross-role rules

├─ safety-agronomic.md

P1 Label claim, dose bounds, PHI, re-entry, resistance, veto conditions

├─ safety-content.md

P1 Category taxonomy, action matrix, six enforcement points

├─ safeguarding-protocol.md

P1 Self-harm path. Clinical/safeguarding review required

├─ privacy-dpdp.md

P1 Consent, retention, deletion, export, audit, media policy

├─ play-compliance.md

P1 Data Safety mapping, permissions rationale, billing, deletion URL

├─ voice-design.md

P1 Mic behaviour, readback, barge-in, replay, failure speech

├─ device-constraints.md

P1 minSdk, RAM, storage, APK budget, image sizing, data/battery budget

├─ i18n.md

P1 Pack structure, script rendering, glossary governance, QA sign-off

├─ eval-strategy.md

P1 Golden sets, labelling workflow, metrics, regression gates

├─ observability.md

P1.5 Tracing, case replay, cost per resolved query

├─ adr/

P1 One file per architectural decision

├─ skills/

Shared capabilities. Agent Skills format: SKILL.md needs YAML frontmatter

├─ README.md

P1 Master index + invariants (renamed from SKILL.md – not a valid skill itself)

├─ CLAUDE.md

P1 Local rules for this folder

├─ _template/

P1 Scaffold: frontmatter, nine sections, references/, prompts/, src/, tests/

├─ speech/

P1 STT + TTS via the Sarvam gateway, code-mix tolerant

├─ translate/

P1 Canonical-language normalisation and render-back

├─ vision/

P1 Crop/pest analysis, quality check, domain gate

├─ retrieval/

P1 RAG over packs, POP, label data, with citation

├─ dose-math/

P1 Area, volume, unit conversion, tank mixing – DETERMINISTIC

├─ confidence/

P1 The single confidence scorer for the whole system

├─ readback/

P1 Spoken numeric confirmation before commit – DETERMINISTIC

├─ uncertainty/

P1 Per-language "not sure" phrasing, human-authored

├─ content-safety/

P1 Classification+ action, all modalities, incl. safeguarding path

- rendering/ - datetime-nlu/ - geo-weather/ - pii-redaction/ - market-data/	- P1 Recommendation card → text, voice, dealer-print - P1 Spoken dates and quantities → structured values - P1 Provider-agnostic forecast normalisation - P1.5 Strip identifiers before storage - P2 Mandi price normalisation
- agents/ - CLAUDE.md _template/ - orchestrator/ - context/ - triage/ - clarification/ - crop-stage/ - weather/ - diagnosis/ - treatment/ - agronomic-safety/ - nutrition/ - escalation/ - outcome/ - irrigation/ - memory/ - variety/ - input-match/ - demand-forecast/ - harvest/ - aggregation/ - market/ - scheme/	One owned decision each. Stateless. Independently evaluable - P1 Local rules - P1 AGENT.md nine-section scaffold + manifest.yaml - P1 Sequences agents, enforces budget - P1 Reads/writes farm graph, assembles case bundle - P1 Routes: cached / simple / full / escalate - P1 Owns the one-missing-question turn - P1 Infers stage, emits stage-due tasks - P1 Forecast → operational window - P1 Ranked hypotheses + confidence. Reference exemplar - P1 Intervention class, dose, timing, method - P1 VETO gate - P1 Stage-wise nutrient schedule - P1 Packages case, routes to expert, tracks SLA - P1 Follow-up scheduling, result capture, ledger write - P1.5 Irrigation scheduling - P1.5 Farmer preference and trust state - P1.5 Pre-sowing variety and seed rate - P1.5 Intervention class → SKU → local stock - P1.5 Predicted input demand - P2 Maturity, pick interval, grading - P2 FPO clustering, pooled demand, harvest windows - P2 Buyer and price signals - P2 Government scheme eligibility
- packs/ - CLAUDE.md - schema/ - crops/ - chilli/ - tomato/ - okra/ - cotton/ - soybean/ - languages/ - te-IN/ ta-IN/ kn-IN/ - mr-IN/ hi-IN/ - en-IN/	Data, never code - P1 Local rules - P1 JSON schema + validator per pack type - P1 Family A - P1 Family A - P1 Family A - P1.5 Family B – resistance + Bt refuge module - P1.5 Family B – weed module - P1 UI strings, agro glossary, profanity lexicon, TTS voice - P1 Pre-login UI default; post-login transcript translate-mode target. UI strings + agro glossary for translate-mode rendering - P1 District variants, local units, seasonal calendars – 5 states
regions/	
- backend/ - CLAUDE.md - api/ - app/ - gateways/ - tests/ - inference/ - app/	Python 3.12 · FastAPI · uv – two Cloud Run services (ADR 0004) - P1 Local rules - P1 Service: orchestration, quota, identity, packs. Own pyproject, uv.lock, Dockerfile - P1 FastAPI app, settings (env only), JSON logging, /healthz /readyz - P1 Provider adapters for this service – every external SDK call lives here, nowhere else - P1 pytest - P1 Service: image analysis. Own scaling profile, pyproject, uv.lock, Dockerfile - P1 FastAPI app, settings (env only), JSON logging, /healthz /readyz

└─ pipeline/	P1 Model-backed analysis, progressively replaced by deterministic CV (ADR 0004)
└─ tests/	P1 pytest
└─ android/ └─ CLAUDE.md └─ app/ └─ core/ └─ network/ └─ database/ └─ sync/ └─ voice/ └─ media/ └─ i18n/ └─ billing/ └─ analytics/ └─ design-system/ └─ feature-onboarding/ └─ feature-farmer/ └─ feature-images/ └─ feature-support/ └─ feature-settings/ └─ feature-billing/ └─ feature-dealer/ └─ feature-fpo/	- Kotlin + Compose, single APK P1 Local rules P1 Application module, DI, role routing, nav host P1 API client, retry, offline-aware P1 Room + outbox pattern P1 WorkManager jobs, backoff, triggers P1 Mic capture, VAD, playback, barge-in, audio focus P1 CameraX, compression, EXIF strip, local queue P1 Language switching, script rendering, audio labels P1 Play Billing v8 client P1 Telemetry, opt-out aware P1 Tokens, typography, day/night, 56dp+ targets, mic dock P1 Language, OTP, consent, permissions, field registration P1 Home, crop timeline, report, diagnosis, ask, history P1 Guided capture, library, picker, quota prompts P1 Global feedback sheet, help, safeguarding surface P1 Language, region, theme, privacy, storage, export, delete P1 Plans and purchase flow P1.5 Demand, leads, counter mode, orders, stock P1 Member list + cluster alerts (read-only); full dashboard P1.5+
└─ consoles/ └─ CLAUDE.md └─ expert/ └─ authoring/ └─ admin/	- Web P1 Local rules P1 Escalation queue, case bundle, approve/edit, SLA – 10 staff P1 Crop and language pack CMS with review and versioning P1 Agent health, evals, cost, quarantine queue, kill switches
└─ evals/ └─ golden/ └─ harness/ └─ safety/ └─ reports/	P1 Per-agent labelled sets, versioned, named labeller P1 Runner, scoring, regression gates P1 Red-team: content, agronomic, multilingual abuse P1 Output history per agent version
└─ ops/ └─ infra/ └─ ci/ └─ runbooks/	P1 IaC, environments P1 Build, test, eval gate, policy lint (see §4) P1 Incident, rollback, safety escalation
└─ legal/	P1 Terms, privacy, content policy, consent copy × 6 languages